FCSG-Net: Frequency-Band Spatial Expert Routing for Lightweight Composite Image Restoration

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by

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Abstract

Archival photographs usually suffer from several degradations at once: optical blur, loss of resolution, sensor or film noise and lossy JPEG compression. Each affects a different part of the image spectrum. This project studies whether a very small restoration network can exploit that structure. The proposed network, FCSG-Net, splits the degraded image into low, mid and high frequency bands with fixed, soft radial masks in the Fourier domain, processes every band with three shared convolutional experts of kernel size 7×7, 5×5 and 3×3, and uses a lightweight gate to select the two most suitable experts for each band at every pixel. The model has 97,088 parameters and costs 9.39 GFLOPs for a 256×256 input. We train FCSG-Net, DnCNN and two FFDNet variants from scratch on the same synthetic composite degradation of DIV2K images, using one schedule and one evaluation protocol, after verifying the implementation and reproducing the published FFDNet result to within 0.01 dB. On 100 validation images FCSG-Net reaches 27.24 dB PSNR, against 27.83 dB for DnCNN and 28.19 dB for blind FFDNet, while using 82.6% fewer parameters than DnCNN. Its low FLOP count does not translate into speed: on a Tesla T4 GPU it is the slowest of the four models. The learned routing is spatially varying and tracks the noise level, with within-image correlations up to 0.81, and overriding the gate's selections at inference costs 0.37 dB on average. Frequency-band routing thus gives a small model inspectable behaviour, but within this budget it does not make the model more accurate or faster than simple convolutional baselines.

Keywords: image restoration, frequency decomposition, mixture of experts, spatial routing, lightweight networks

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We also thank the authors of DIV2K for making their dataset public, and Kai Zhang and colleagues for releasing the FFDNet code and pretrained weights that let us check our implementation against a published result.

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Chapter 1

Introduction

1.1 Motivation and Problem

Old photographs rarely suffer from a single corruption. A print scanned at modest resolution, stored as a compressed JPEG and taken through a soft lens on grainy film carries all of those defects at once, and each affects a different part of the spectrum. Blur removes high frequencies, resampling discards detail above a resolution limit, noise dominates wherever the signal is weak (in natural images, the high frequencies), and JPEG quantises the mid and high frequencies of each 8×8 block. Most restoration networks treat the mixture as one problem, and the strongest of them need tens of millions of parameters.

This project asks whether a very small network can use that spectral structure explicitly: split the image into frequency bands, and let a learned gate choose, for every band and every pixel, between convolutional experts of different kernel sizes. We call the network FCSG-Net.

Formally, a clean RGB image $f \in [0,1]^{H \times W \times 3}$ is degraded as $x = \mathcal{D}(f;\theta)$ with $\theta = (\sigma_{\text{blur}}, s, \sigma_n, q)$: Gaussian blur, bicubic down- and up-sampling by s, Gaussian noise of standard deviation σ_n, and JPEG at quality q. The parameters are random per image and hidden from the model. We seek $\hat{f} = g_\phi(x)$ close to f in PSNR, SSIM and LPIPS, with fewer than five million parameters and fewer than ten GFLOPs for a 256×256 input.

1.2 Objectives and Contributions

We test the hypothesis that *spatial top-2 routing over shared kernel-size experts, applied after a fixed frequency decomposition, yields a routing policy that varies with content and degradation and that the trained network relies on, within the budget.* The objectives are:

O1 build FCSG-Net and verify it before training;

O2 establish trustworthy baselines by reproducing a published FFDNet result;

O3 compare FCSG-Net, DnCNN and two FFDNet variants trained under one matched protocol, in quality, size and measured speed;

O4 characterise the learned routing.

The contributions are a 97,088-parameter network with its verification suite, a matched composite-degradation benchmark, a routing analysis linking expert weights to the recorded degradation, and a plain account of the outcome: FCSG-Net is within 0.6 dB of DnCNN with 5.7 times fewer parameters, but it is not faster on a GPU. We make no state-of-the-art or novelty claim; frequency decomposition and expert routing are both established (Chapter 2).

Chapter 2 reviews related work, Chapter 3 describes the method, Chapter 4 the experimental setup and verification, Chapter 5 the results and routing analysis, and Chapter 6 concludes.

Chapter 2

Related Work

Published scores from the works below use other tasks, datasets or noise protocols, so they serve only as context. Our result tables contain only models we trained and evaluated ourselves.

CNN restoration. DnCNN [1] showed that a deep stack of convolutions with batch normalisation learns denoising much more easily by predicting the noise residual. FFDNet [2] processes four half-resolution sub-images and takes a noise-level map as an extra input, so one network covers a range of noise strengths; the author's colour model has 852,108 parameters and reports 31.21 dB on CBSD68 at $\sigma = 25$. Its map describes only Gaussian noise, not blur or JPEG. Larger transformer-based restorers reach higher quality but need tens of millions of parameters, far outside our budget.

Frequency-domain restoration. MWCNN [3] replaces pooling in a U-Net with wavelet transforms and processes subbands jointly, which motivates our cross-band fusion. The closest work is SFNet [4], which already performs local, content-adaptive frequency selection inside a U-shaped network. Our distinction is narrower: FCSG-Net uses *fixed* radial masks and puts the learned spatial decision in the choice among shared 7×7, 5×5 and 3×3 experts, which we can then compare with known degradation parameters.

Mixture of experts. Shazeer et al. [5] route each example to a few experts with a top-k gate and balance load with auxiliary penalties. The efficiency argument for sparse experts depends on not running the unselected ones. Our implementation computes every expert and mixes them with sparse weights, so sparsity here aids analysis, not speed.

Metrics. PSNR is $10 \log_{10}(1/\mathrm{MSE})$ on $[0, 1]$ RGB images, computed per image and averaged. SSIM compares local means, variances and covariances in an 11×11

Gaussian window ($\sigma = 1.5$). LPIPS [6] measures distance between deep features (AlexNet variant); lower is better, and it can disagree with PSNR.

Table 2.1 summarises the positioning. Each ingredient of FCSG-Net is established; the project studies one combination of them under a tight budget.

Table 2.1: Positioning of FCSG-Net against closely related methods.

Method	Freq. split	Learned selection	Spatial	Notes
DnCNN [1]	–	–	–	residual CNN
FFDNet [2]	–	–	–	noise-map input
MWCNN [3]	wavelet	–	–	joint subbands
SFNet [4]	learned	frequency	local	closest prior work
Sparse MoE [5]	–	experts	–	per-example routing
FCSG-Net (ours)	radial FFT	experts, top-2	per pixel	97k parameters

Chapter 3

Proposed Method: FCSG-Net

3.1 Degradation Model

The synthetic pipeline applies four operations in physical order (Table 3.1). The image is quantised to 8 bits before blur and before JPEG encoding. Each sample stores its sampled parameters, which the network never sees but which we use to analyse the routing in Section 5.3. The mean input PSNR on our validation set is 20.99 dB.

Table 3.1: Composite degradation, sampled independently for every image.

Stage	Parameter	Distribution
Gaussian blur	σ_{blur}	$\mathcal{U}(0.5, 2.0)$ pixels
Down/up resampling	scale s	$\mathcal{U}(1.0, 2.0)$, bicubic
Additive noise	σ_n	$\mathcal{U}(5, 50)/255$, clipped
JPEG compression	quality q	integer in $\{30, \dots, 95\}$, 4:4:4

3.2 Architecture

The input is split into low, mid and high frequency bands, each an ordinary RGB image. Three shared experts E_7, E_5, E_3 process every band, and a spatial gate gives each expert a weight per band and pixel, keeping the top two (Figure 3.1):

$$y_b(u,v) = \textstyle\sum_e w_{b,e}(u,v)\, E_e(x_b)(u,v), \qquad b \in \{L, M, H\}, \tag{3.1}$$

$$\hat{f} = x + \text{Refine}\big(\text{SEFusion}([y_L, y_M, y_H])\big). \tag{3.2}$$

Frequency decomposition. With X the orthonormal real FFT of x and r the radial frequency in cycles per pixel, a soft low-pass mask of cutoff c and width $\Delta = 0.05$ is $\text{LP}_c(r) = \frac{1}{2}\big(1 + \cos(\pi\,\text{clip}((r-c)/\Delta + \frac{1}{2}, 0, 1))\big)$. With $r_1 = 0.10$ and $r_2 = 0.25$

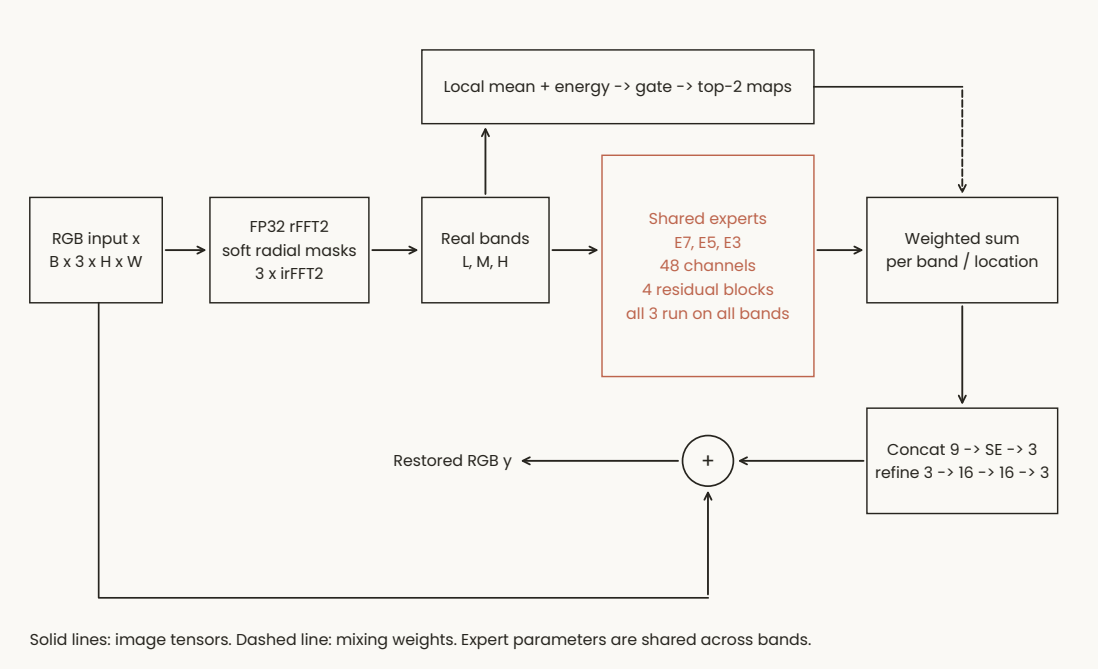


Figure 3.1: FCSG-Net data flow. Expert parameters are shared across bands.

the band masks are $M_L = \mathrm{LP}_{r_1}$, $M_M = \mathrm{LP}_{r_2} - \mathrm{LP}_{r_1}$ and $M_H = 1 - \mathrm{LP}_{r_2}$, and $x_b = \mathcal{F}^{-1}(M_b X)$. Three choices matter. The masks sum to one, so $x_L + x_M + x_H = x$ exactly. Cutoffs are normalised frequencies rather than FFT bin indices, so a band means the same thing on a 128×128 training crop as on a full image. And each band returns to the spatial domain before any learning: a convolution over a spectrum has no clear meaning and complex inputs make training unstable, so the experts stay plain real-valued CNNs.

Experts. Each expert applies a 2×2 pixel unshuffle (12 channels at half resolution), a 1×1 convolution to 48 channels, four residual blocks of two depthwise-separable stages with kernel $k \in \{7, 5, 3\}$, a 1×1 tail added to the unshuffled input, and a pixel shuffle back. Separable, half-resolution processing is what keeps nine expert passes inside the compute budget. We name the experts only by kernel size, since we have no evidence about what each one specialises in.

Spatial top-2 gate. A per-image gate would be a lookup table that tends to settle on a constant choice. Our gate pools the mean and mean of squares of the nine band channels in 16×16 windows, maps these 18 statistics through a $18 \to 24 \to 9$ network to one logit per band and expert, and upsamples the logits to full resolution. At each pixel the two largest logits per band are kept and softmax-normalised; the third expert gets zero weight. The dense softmax $p_{b,e}$ over all three is kept for the loss. The gate has 681 parameters. All experts are still computed before mixing.

Fusion and refinement. The nine band channels pass through a squeeze-and-excitation block, a 1×1 projection to RGB and three 3×3 convolutions ($3 \to 16 \to 16 \to 3$). The output is a residual added to the input, as in DnCNN.

3.3 Training Objective

$$\mathcal{L} = \mathcal{L}_{\text{char}} + 0.05\,\mathcal{L}_{\text{freq}} + \lambda_{\text{ent}}(t)\,\mathcal{L}_{\text{ent}}. \tag{3.3}$$

$\mathcal{L}_{\text{char}} = \text{mean}\sqrt{(\hat{f} - f)^2 + \varepsilon^2}$, $\varepsilon = 10^{-3}$, is the Charbonnier loss. $\mathcal{L}_{\text{freq}}$ is the mean absolute FFT of the error, which weights all frequencies equally. $\mathcal{L}_{\text{ent}} = \sum_b \text{mean} \sum_e p_{b,e} \log p_{b,e}$ is negative routing entropy; minimising it keeps all experts in use early on. Its coefficient decays linearly from 0.01 to 0 so that the gate is free to commit by the end. The baselines use the same loss without the entropy term.

3.4 Size and Compute

FCSG-Net has 97,088 parameters: 39,228, 30,012 and 23,868 in the 7×7, 5×5 and 3×3 experts, 681 in the gate, 96 in fusion and 3,203 in refinement. We count each multiply-accumulate as two FLOPs and add an estimate for the FFTs; a 256×256 pass costs 4.66 GMACs, or 9.39 GFLOPs, inside the limit. Section 5.2 shows this count was a poor predictor of running time.

Chapter 4

Experimental Setup

4.1 Data, Baselines and Training

We use DIV2K [7]: 800 training images and 100 validation images, the latter used for every reported result. Training uses a cache of 32 fixed 128×128 crops per training image (25,600 tiles), with random flips and rotations, and applies the degradation on the fly, so every epoch sees new corruptions. A fixed set of 5,000 pairs (seed 1234) was also exported for reproducibility.

The baselines are an RGB DnCNN (17 layers, width 64, 558,403 parameters) and FFDNet in the author's colour configuration (852,108 parameters) in two variants. *Oracle* FFDNet receives the true sampled noise level, which no real user would have, so it is an upper reference. *Blind* FFDNet receives a constant map of 27.5/255 and sees exactly what the other models see. All four models are trained from scratch with the same settings (Table 4.1). Each model was trained once with an unfixed seed, on a Tesla T4 in Kaggle sessions that resume from checkpoints.

Table 4.1: Training configuration, identical for every method.

Setting	Value
Optimiser and schedule	AdamW, 2×10^{-4} cosine decay to 10^{-6}
Updates and batch	200,000; 16 crops of 128×128
Loss	$\mathcal{L}_{\text{char}} + 0.05\,\mathcal{L}_{\text{freq}}$ (+ entropy for FCSG-Net)
Precision	FP16 mixed precision
Hardware	NVIDIA Tesla T4 (Kaggle)

4.2 Evaluation Protocol

Each of the 100 validation images is degraded once with a seed equal to its index, so all models see identical inputs. Inference uses 256×256 tiles with 32-pixel overlap,

averaged where tiles overlap. Training-time validation on 16 fixed crops is used only for monitoring; it is not comparable with full-image results (30.59 against 27.24 dB for FCSG-Net).

4.3 Verification and Baseline Reproduction

Before training, a notebook on the same T4 ran the checks in Table 4.2 and stopped at any failure. The overfit check is the most useful: a model that cannot memorise one image in 200 steps has a bug, and it is better to learn that in seconds than after ten hours of training.

Table 4.2: Pre-training verification checks.

Check	Criterion	Measured
FFDNet weights, CBSD68 σ=25	within 0.5 dB of 31.21	31.220 dB
Band reconstruction	$x_L+x_M+x_H = x$	max error 6.0×10^{-7}
Parameters / GFLOPs at 256^2	< 5M / < 10	97,088 / 9.39
Memory, forward + backward	< 4 GB	0.60 GB
One-image overfit, 200 steps	MSE < 0.01, $< \frac{1}{4}$ initial	ratio 0.18
Checkpoint restore, reproducible data	identical	passed

To validate the FFDNet code against a published number we evaluated the author's pretrained colour weights [2] on the 68 CBSD68 images with the author's protocol, obtaining 31.220 dB against 31.21 dB in the paper, a difference of 0.01 dB. As an end-to-end check of the training loop, DnCNN trained on clipped Gaussian noise at $\sigma = 25$ reached 32.65 dB on DIV2K validation and 31.04 dB on CBSD68.

Chapter 5

Results and Analysis

5.1 Restoration Quality

Table 5.1 gives the main result on the 100 validation images.

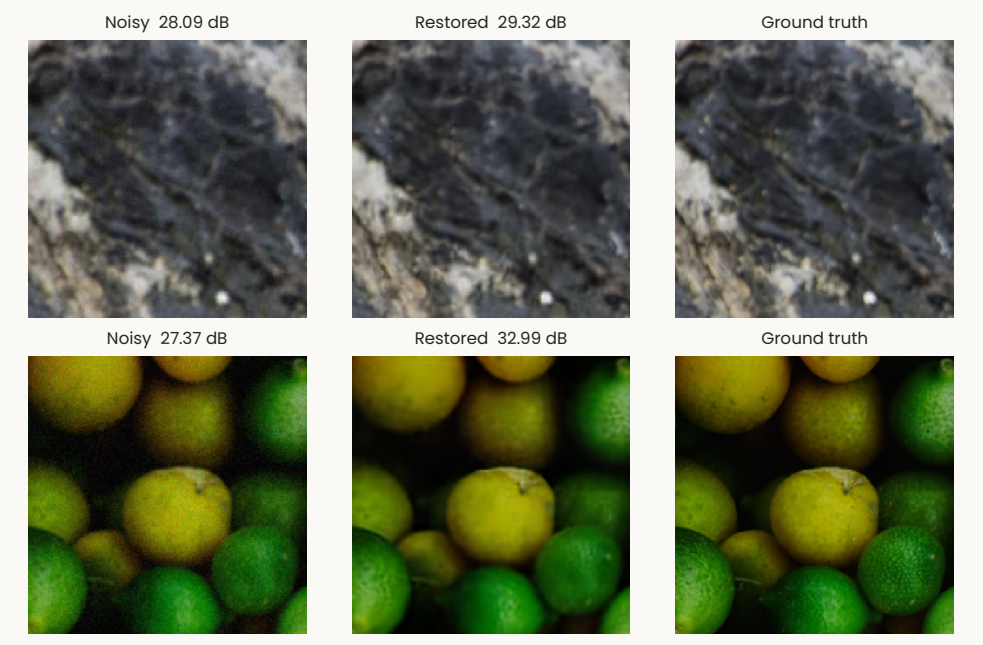
Table 5.1: Composite restoration on 100 DIV2K validation images after 200,000 updates (input PSNR 20.992 dB). GFLOPs at 256×256.

Method	PSNR↑	SSIM↑	LPIPS↓	Params	GFLOPs
DnCNN	27.831	0.7795	0.3242	558,403	73.43
FFDNet, blind	28.190	0.7899	0.3176	852,108	27.89
FFDNet, oracle†	28.240	0.7920	0.3162	852,108	27.89
FCSG-Net	27.243	0.7529	0.3961	97,088	9.39

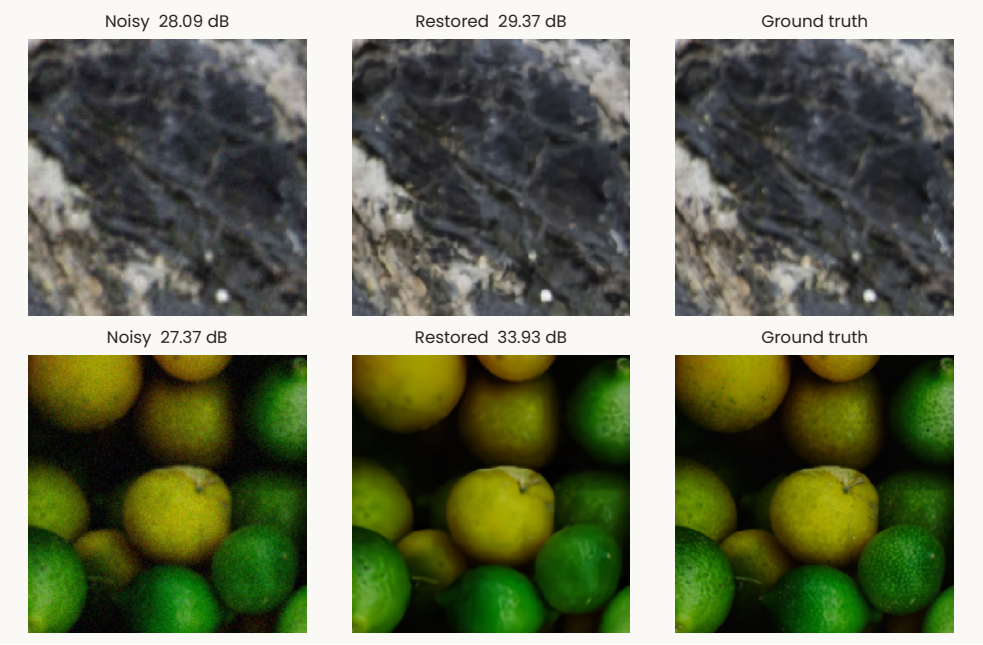
†Given the true noise level; an upper reference.

FCSG-Net improves the input by 6.25 dB but is the weakest model on all three metrics: 0.59 dB behind DnCNN and 0.95 dB behind blind FFDNet, with a proportionally larger gap in LPIPS. It does this with 82.6% fewer parameters and 87.2% fewer reported FLOPs than DnCNN, and 88.6% fewer parameters and 66.3% fewer FLOPs than FFDNet. The deficit is systematic rather than driven by a few images: both FFDNet variants beat FCSG-Net on all 100 images, and DnCNN beats it on 93. The six images where FCSG-Net wins against DnCNN, by up to 2.07 dB, show that the two designs fail in different places.

Blind FFDNet is the strongest fair model even though its noise map is constant, so its advantage comes from the architecture rather than from noise conditioning. The oracle map adds only 0.05 dB, because noise is one of four corruptions. Validation curves had flattened for all models by 200,000 updates, so we have no evidence that longer training would close the gap. Figure 5.1 shows an example restoration.



(a) FCSG-Net



(b) FFDNet, blind

Figure 5.1: Degraded input, restoration and clean target on validation images 0801 and 0802 (top-left 320×320 region; PSNR is for the whole image).

5.2 Computational Cost

Table 5.2 reports median FP32 forward latency at batch 1 on a Tesla T4 after warm-up.

Table 5.2: Median forward latency (ms) and peak inference memory at 256^2 (Tesla T4, FP32).

Method	128^2	256^2	512^2	Peak MB
FCSG-Net	5.10	19.40	75.59	51.3
DnCNN	4.22	18.04	72.47	37.8
FFDNet, blind	1.69	6.30	24.12	20.2

Despite needing an eighth of DnCNN's FLOPs and a third of FFDNet's, FCSG-Net is the slowest model at every size and uses the most memory. Training showed the same pattern: 5.84 steps per second against 9.41 for DnCNN and 26.67 for FFDNet. We did not profile individual operators, so the explanation is our reading of the architecture: nine full expert passes per image, depthwise convolutions whose speed is limited by memory traffic rather than arithmetic, many small layers with launch overhead, and the FFTs. FFDNet runs one stack of dense 3×3 convolutions, which GPUs execute very efficiently. The FLOP budget set at the start was therefore the wrong proxy for speed on this hardware (Figure 5.2). Tiling is not the cause of the quality gap: changing the tile size from 256 to 512 pixels changes FCSG-Net by only 0.013 dB on a ten-image subset.

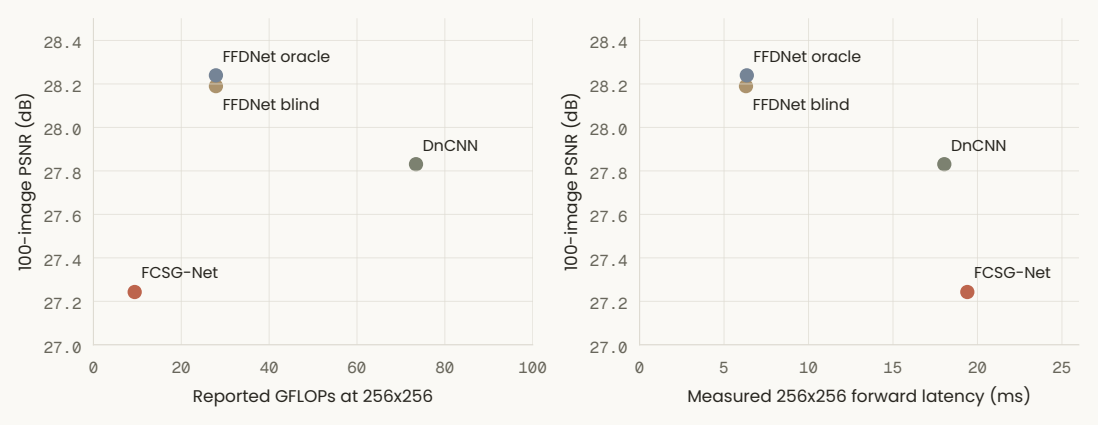


Figure 5.2: Validation PSNR against reported GFLOPs (left) and measured 256×256 latency (right).

5.3 Routing Analysis

We analysed the final FCSG-Net checkpoint on ten validation images, each degraded with eight seeds (80 samples, ten independent images).

Where the gate routes. The dense routing probabilities end close to uniform (mean entropy 1.09 to 1.10 nats per band against a maximum of 1.10), which alone would suggest the gate learned nothing. The top-2 selection says otherwise. In the high band the 3×3 expert is selected at 99.98% of pixels, with the second slot shared between the larger experts depending on location. The maps follow image structure (Figure 5.3), and because top-2 selection switches when two logits swap rank, nearly uniform probabilities can still give a consistent and sharp choice.

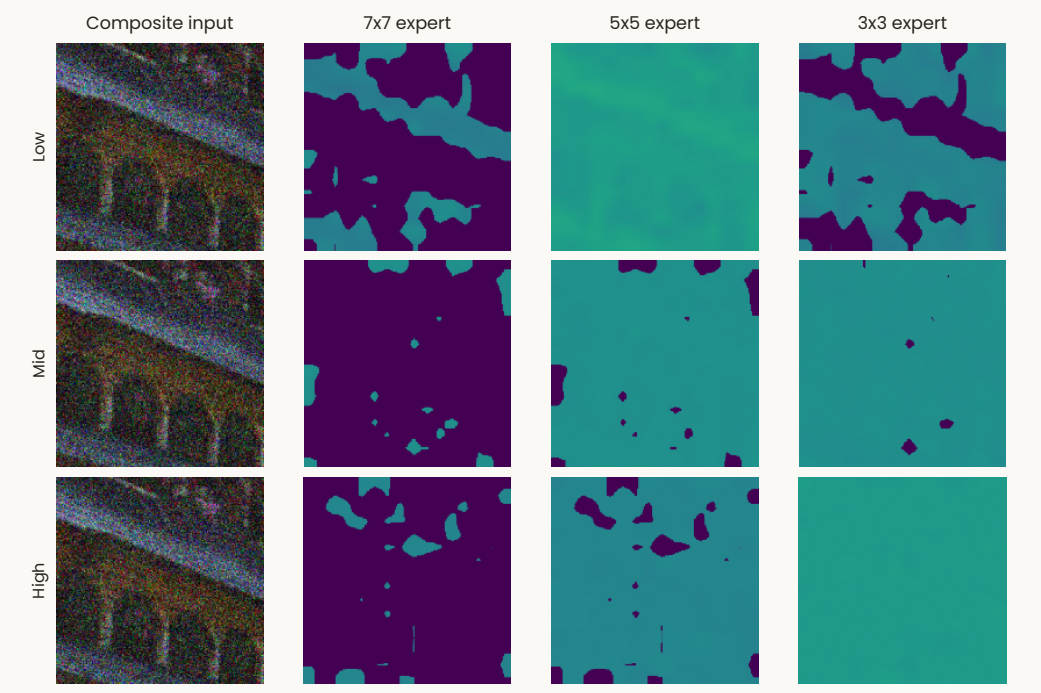


Figure 5.3: Expert weights for validation image 0834. Rows: low, mid, high band. Columns: degraded crop and weights of the 7×7, 5×5 and 3×3 experts.

Routing versus degradation. After removing each image's mean, the high-band 3×3 weight correlates at $r = +0.81$ with the noise level, the low-band 5×5 weight at $+0.80$ and the low-band 7×7 weight at -0.73: with more noise, the gate moves weight towards smaller kernels. The six strongest correlations are all with noise; the strongest with blur is $+0.42$ and with JPEG quality -0.39. These are exploratory results from ten images and 36 correlations without multiple-comparison correction, and the four parameters vary together, so they show a consistent response but not a causal one.

Does the model need the gate? Keeping the trained weights and overriding the gate at inference gives the changes in Table 5.3.

Flattening the weights within the chosen pair barely matters, while removing the choice itself costs 0.37 dB on average and nearly 1 dB on one image. The trained

Table 5.3: PSNR change when the trained gate is overridden at inference (10 images).

Intervention	Mean ΔPSNR	Min / max
Equal weights on the selected pair	−0.040 dB	−0.137 / +0.028
Uniform weights over all three experts	−0.372 dB	−0.986 / +0.163

network depends on *which* experts are combined far more than on how they are weighted. This shows the gate does real work in this model; whether a model trained without a gate would be worse needs a matched training ablation, which we leave to future work.

5.4 Discussion

The routing works in the sense we set out to test: it varies spatially, tracks the noise level and matters to the trained model. It does not make the model competitive. With under 100,000 parameters, FCSG-Net trails baselines six to nine times larger by 0.6 to 1.0 dB, most likely for lack of capacity, and on a GPU it is not faster. Frequency-band routing is a cheap way to make a small model's behaviour inspectable; within our budget it is not by itself a way to make it better or faster.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

FCSG-Net splits an image into three frequency bands and routes each band, per pixel, to two of three shared convolutional experts. With 97,088 parameters and 9.39 GFLOPs it passed every pre-training check (O1), and our FFDNet code reproduced the published CBSD68 result to within 0.01 dB (O2). Under one matched composite-degradation protocol it reached 27.24 dB, 0.59 dB below DnCNN and 0.95 dB below blind FFDNet, with 82.6% and 88.6% fewer parameters respectively, yet it was the slowest model on a Tesla T4 (O3). Its gate varies spatially, tracks the noise level with correlations up to 0.81, and the trained model loses 0.37 dB when its selections are overridden (O4). Frequency-band routing gave a small model inspectable behaviour, but not better accuracy or speed than simple baselines.

6.2 Limitations

Each model was trained once, and all results are on the validation split with no untouched test set. The degradation is synthetic; we did not test on real archival photographs or run a human study. The routing analysis uses ten images and exploratory correlations, and the interventions measure the trained model's dependence on its gate rather than the value of having one. Latency was measured on one GPU without an operator profile. All experts are always computed, so sparse routing saves no computation.

6.3 Future Work

The most important next step is a matched training ablation comparing learned routing against fixed uniform weights from the same initialisation, with several seeds,

followed by ablations of the kernel sizes, a per-image gate, plain concatenation instead of SE fusion, and hard masks. Since the model relies mainly on which experts it picks, skipping unselected experts could cut up to a third of the expert cost, but an operator profile should come first, along with latency measurements on CPU and mobile hardware. Widening the experts within the parameter budget, which is fifty times larger than the model uses, would address the capacity gap if cheaper expert evaluation frees compute. Finally, any claim about archival restoration needs real archival images, a controlled degradation sweep and a held-out test set.

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